

Station 1: Macromolecules and Organelles

Directions: Macromolecules are extremely important to how our cells function. Use the **chart** below and your knowledge of macromolecules from last unit to answer the questions under station 1 in your lab packet.

	Carbohydrates	Lipids	Proteins	Nucleic Acids
Monomer	Monosaccharide	Fatty acid	Amino acid	Nucleotide
Polymer	Polysaccharide	Triglycerides	Polypeptides	DNA/RNA
Function	Quick energy	Long term energy/insulation/hormones	Structure, transport, signaling, etc. (LOTS of functions)	Store genetic information used to make proteins
Example	Glucose, Lactose	Testosterone, estrogen, oil, wax	Hemoglobin, insulin, myosin, enzymes	DNA, tRNA, mRNA

Station 2: Human Cells

Many of the cells in our bodies do not look like the typical animal cell shown in biology textbooks. Differences in cell structure are related to the difference in the function of different types of cells.

Directions: Use the information below to help you answer questions about the different animal cells in humans.

Blood Cells:

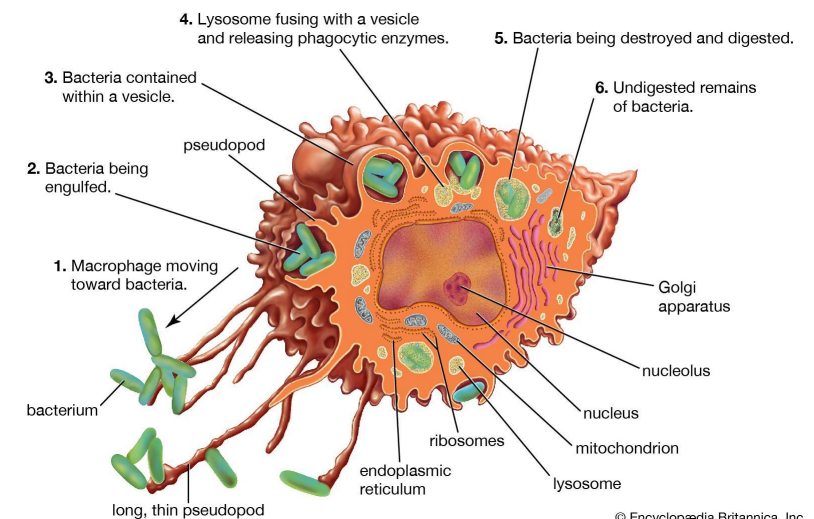
A factor that plays a part in cell function is how much of specific type of organelle is found in that cell. An example of this is red blood cells in mammals. Red blood cells have no nucleus, ribosomes, or mitochondria. Instead, they are filled with hemoglobin which is a protein that helps to carry oxygen. This large amount of oxygen is needed for homeostasis in mammals. When a red blood cell is damaged or starts to carry less oxygen, the whole cell is destroyed. Your red blood cells are replaced about every four months.

A different type of blood cell called a lymphocyte, or white blood cell, are responsible for immune response in the body. They have a variety of functions including recognizing infected cells, signaling that there are pathogens present, and destroying pathogens like bacteria (macrophages)

Red blood cells



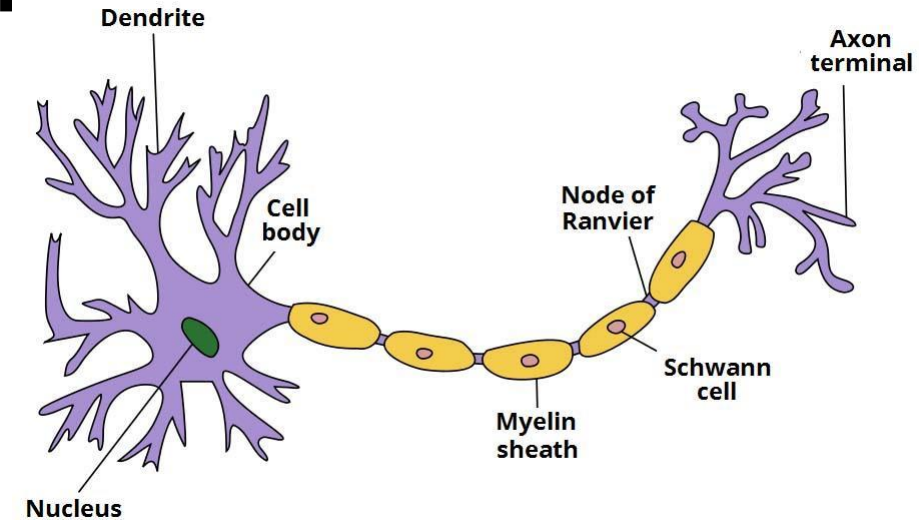
Macrophage



Station 2: Human Cells Cont

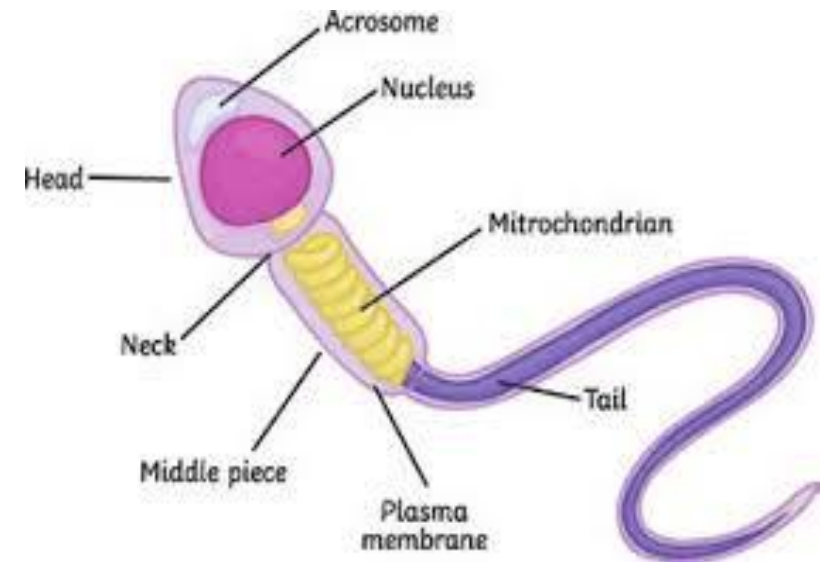
Nerve Cells:

This shows nerve cells found in the human brain. Nerve cells have a thick main body and many long extensions. These long fibers help the nerve cell connect with hundreds of other nerve cells that control our behavior. They also have a fatty covering called the myelin sheath which aids in signal conduction.



Sperm Cells:

Sperm cells have a very specialized structure, including a flagellum, and very little cytoplasm. Sperm cells function to find a female egg in order to fertilize it.

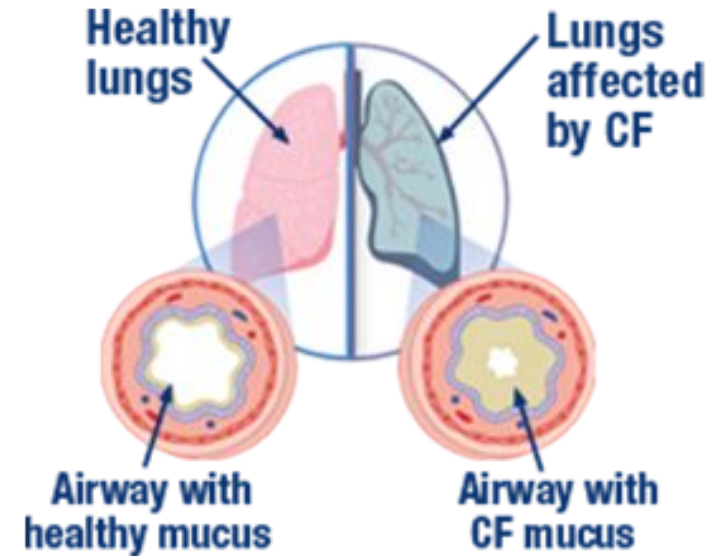


Station 3: Organelles and Disorders

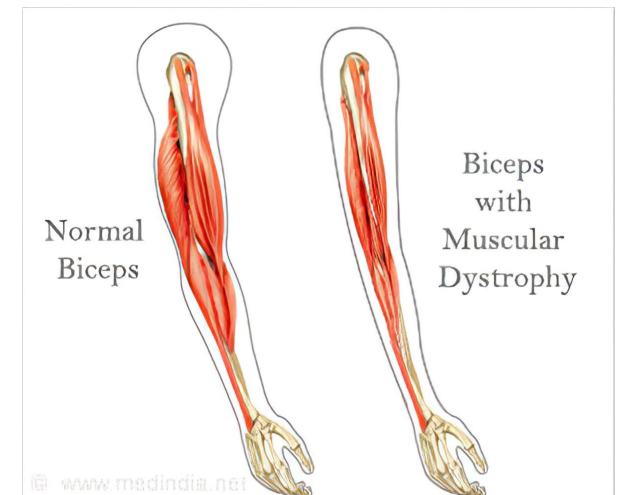
Genetic disorders often result from an organelle in cells not being able to do its job properly as a result of mutations in DNA.

Directions: Read the scenarios below to see what happens when an organelle does not work correctly and answer the questions in your packet.

Cystic Fibrosis: A child is brought to the doctor with a chronic cough. The doctors run tests and find that the child has a lot of salty mucus built up in her lungs. The child's doctor concludes that the child's cells are having trouble making proteins that control the movement of salt and water into and out of the cell.



Muscular Dystrophy: A young man becomes very sick and his doctor does a series of tests. The man has damage to his nervous system, heart irregularities, and worsening muscle weakness. The doctor knows that these organs are especially are very sensitive to cellular energy levels and need the cells to produce a lot of energy in order to function correctly.



Station 3: Organelles and Disorders Cont.

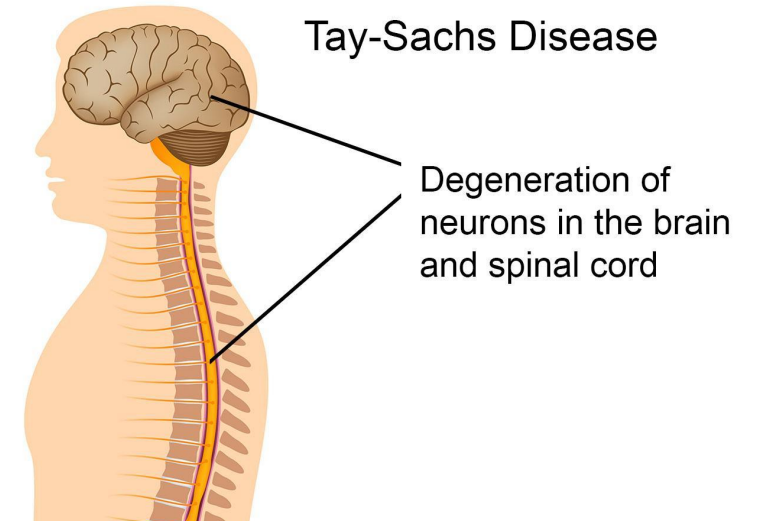
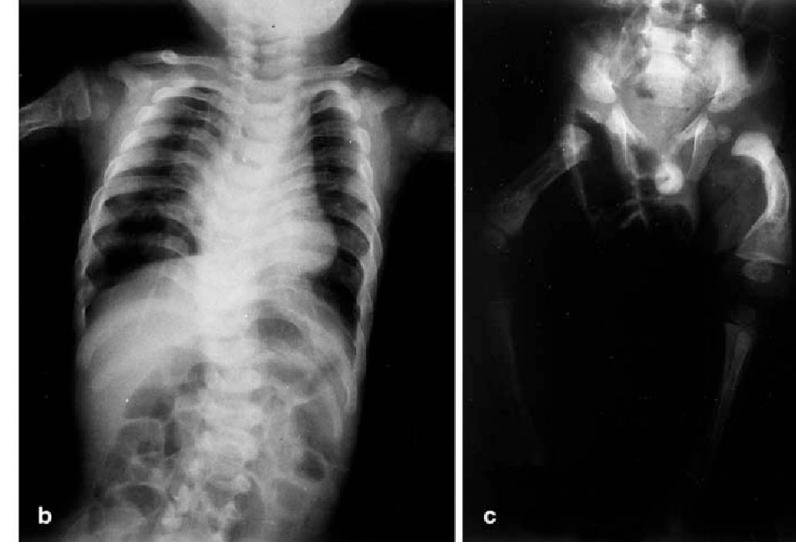
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I Cell Disease

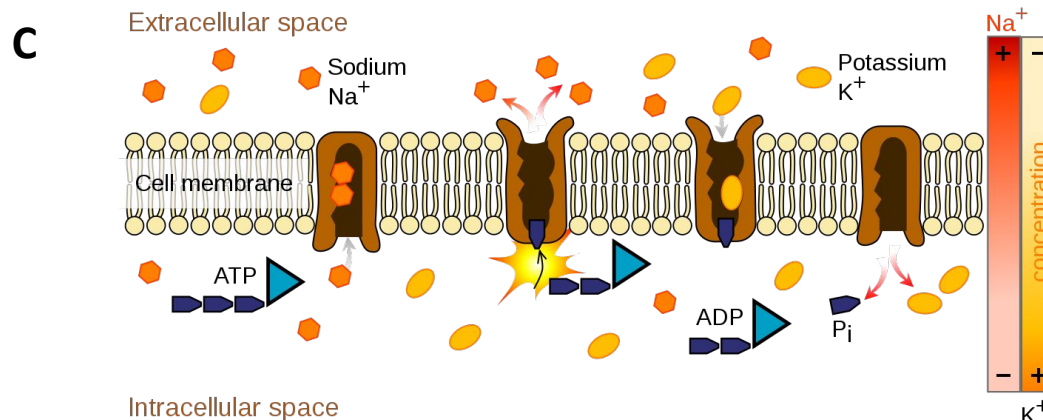
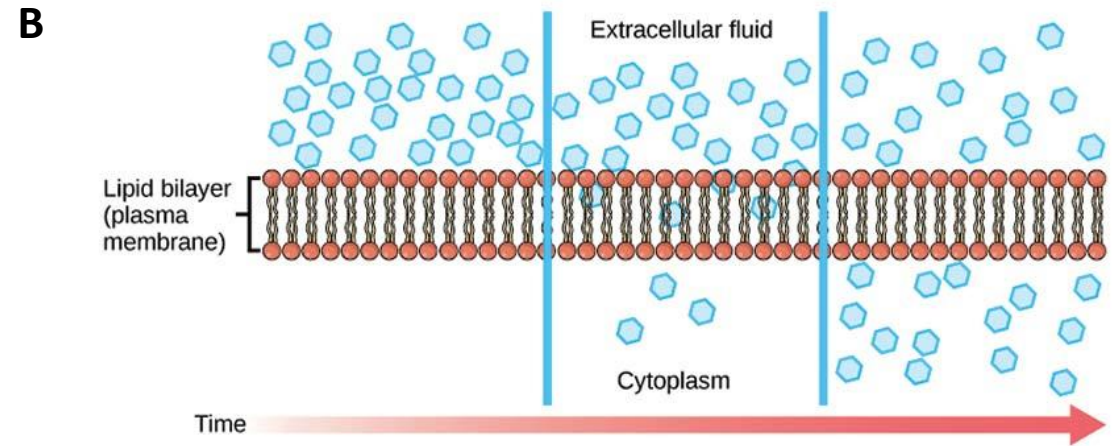
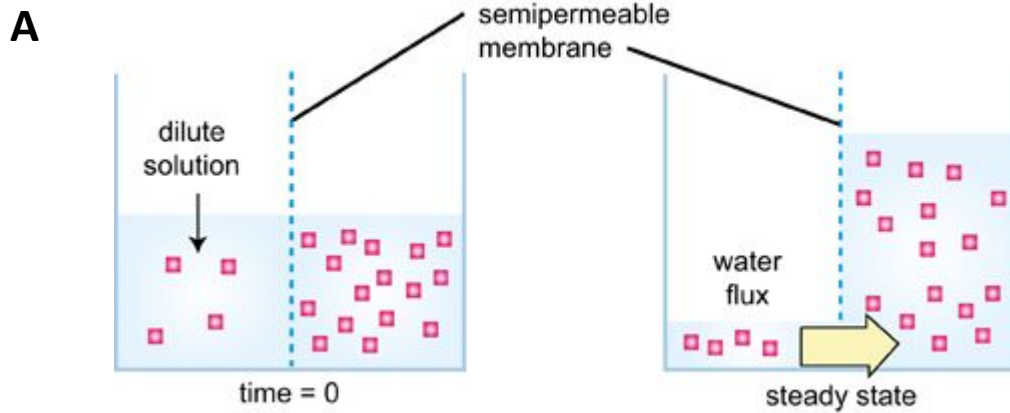
A baby is born with skeletal abnormalities, distinctive facial features, and is failing to gain weight. The doctor runs genetic tests, and finds the baby's cells are not able to finish and sort proteins correctly. As a result, proteins are not shipped to the right locations.

Tay Sach's Disease: A doctor studies cells taken from the spinal fluid of a very sick baby. The doctor finds a buildup of a toxic substance called GM2 ganglioside. The toxic substance is normally broken down in healthy cells. When this substance is not broken down, it builds up and causes dysfunction of brain cells which can cause mental disabilities and nervous system problems.



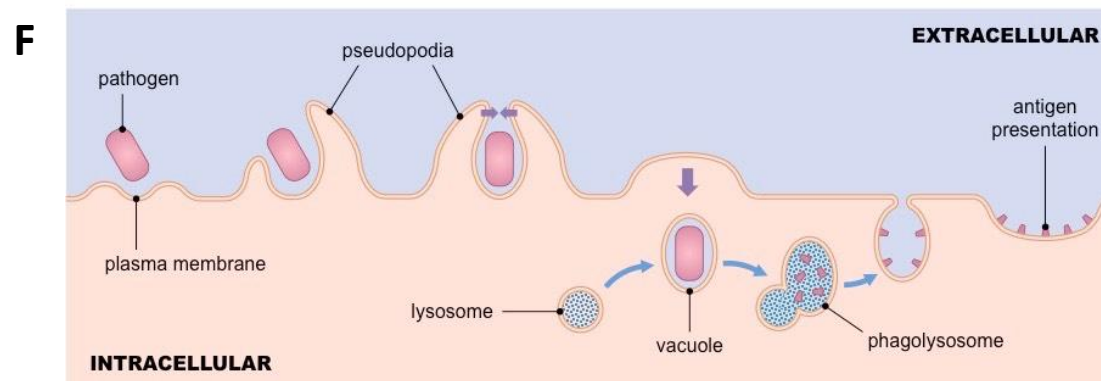
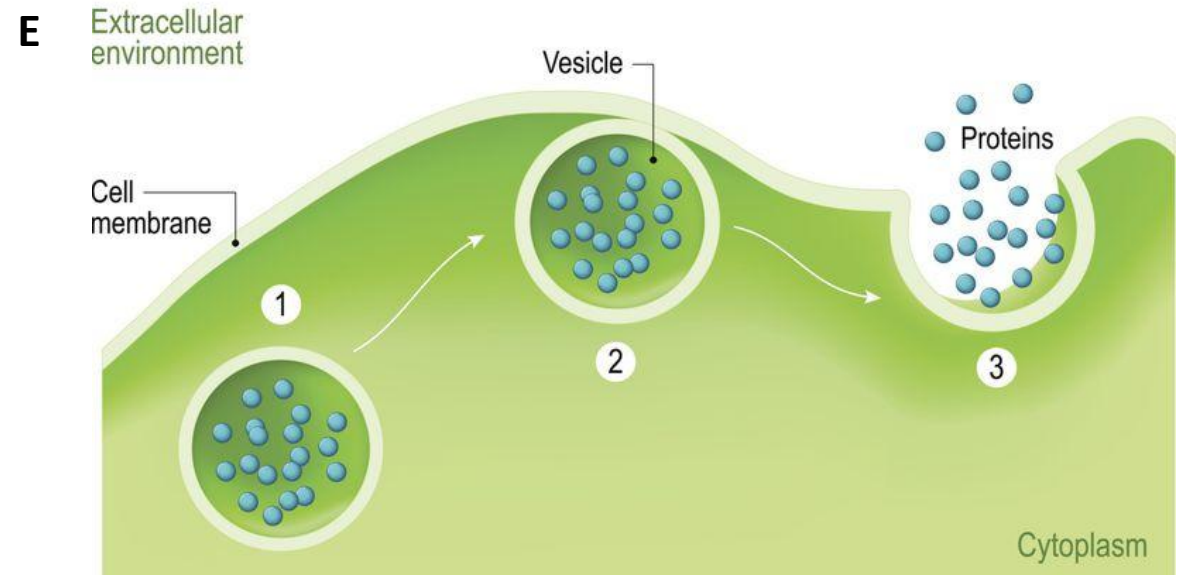
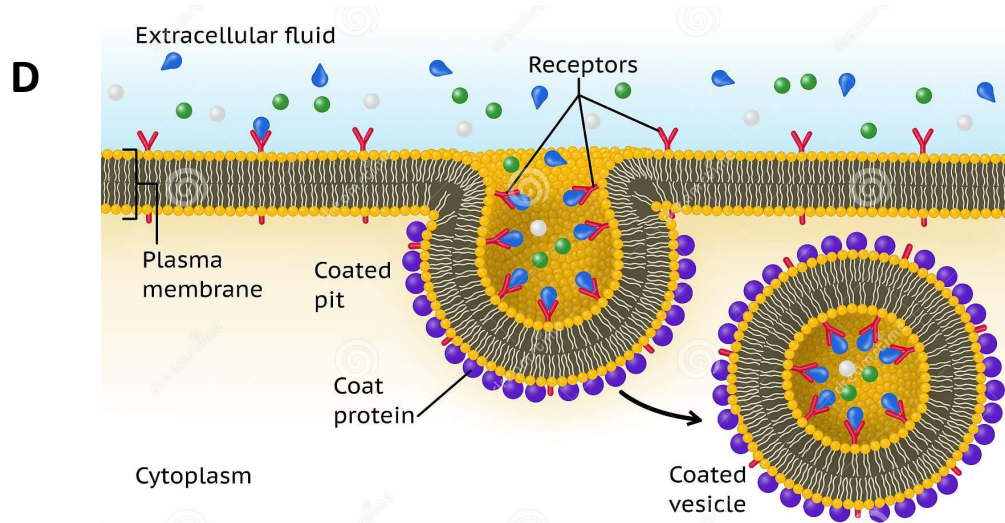
Station 4: Identifying Types of Transport

For each of the diagrams below- decide which type of transport is being shown and state whether it is active or passive.



Station 4: Identifying Types of Transport Cont.

For each of the diagrams below- decide which type of transport is being shown and state whether it is active or passive.



Station 5: oddballs of the natural world- Protists

We often think of Eukaryotes as large complex organisms like humans, animals, and plants. However, there are many Eukaryotes that are both microscopic and single-celled. A diverse group called Protists are single celled organisms that often defy classification. Many are like animals and eat to obtain nutrients while others can utilize photosynthesis.

Directions: Examine the samples of protists on the microscope or picture and answer the questions on your handout.

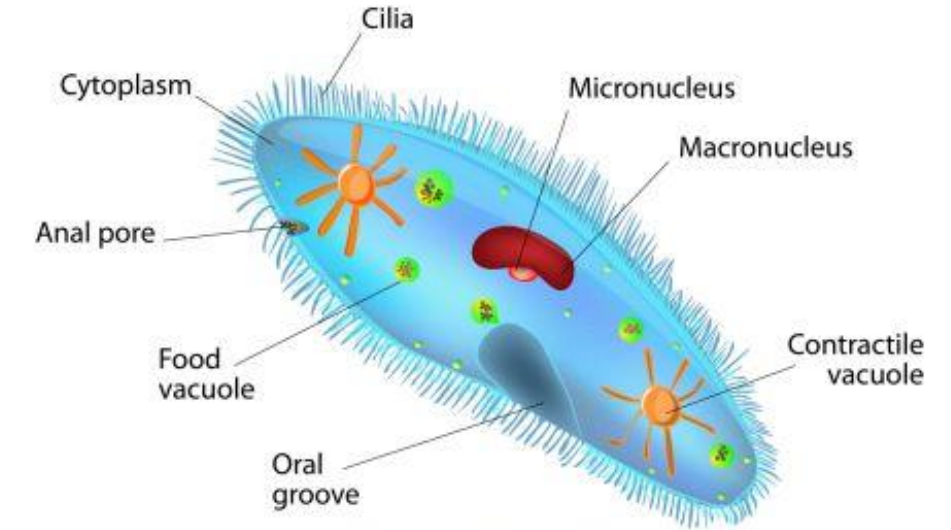
Paramecium

This organism is a single celled protists found in freshwater. It is covered in cilia which are used to bring food closer to it. It can also use these cilia to propel itself in water. As it swims, it captures food in its mouth.

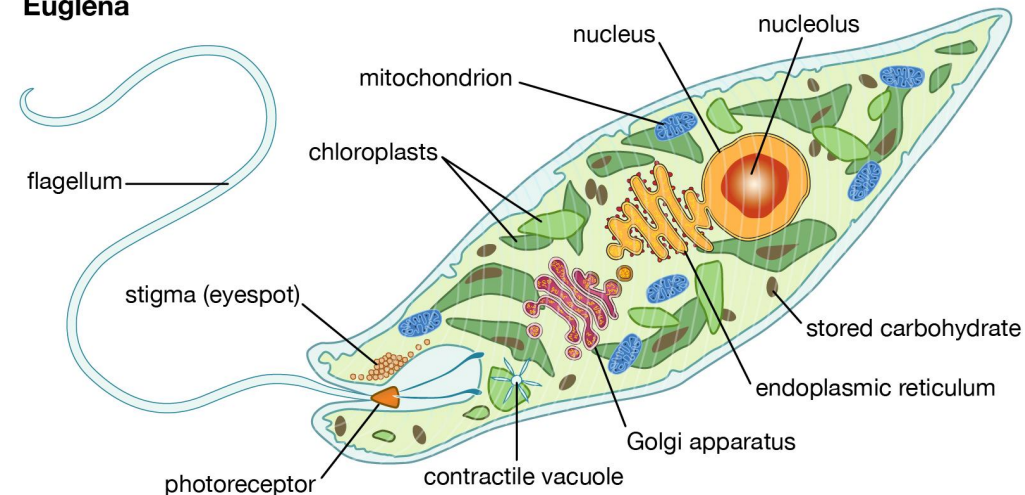
Euglena

Euglena are also found in freshwater and single celled. They spend the daytime performing photosynthesis using their many chloroplasts while hunting for prey at night. Many have flagella used for movement. Although it does not have a cell wall it does have a large vacuole used to regulate its water balance.

Paramecium



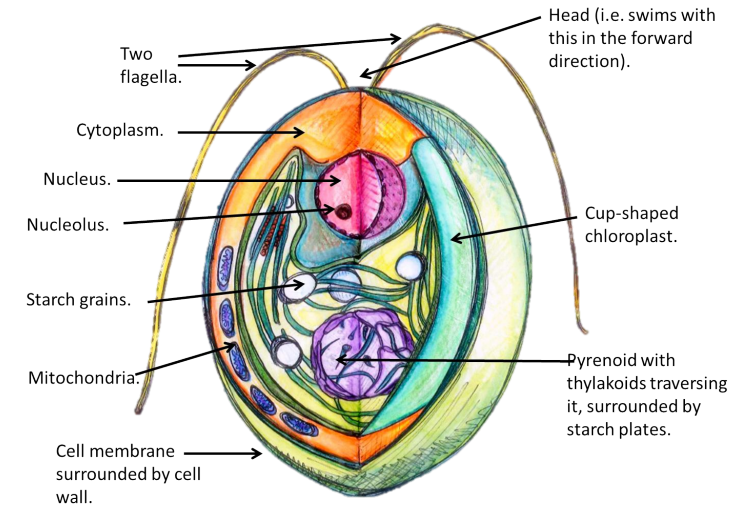
Euglena



Station 5: oddballs of the natural world- Protists

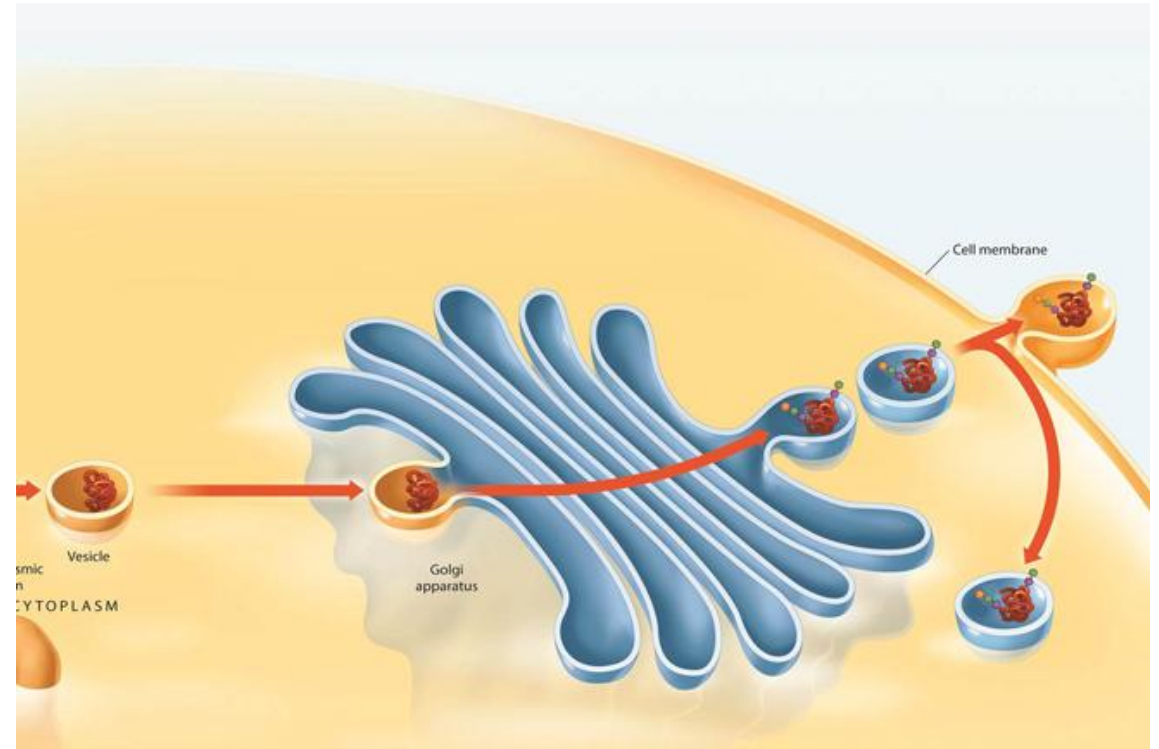
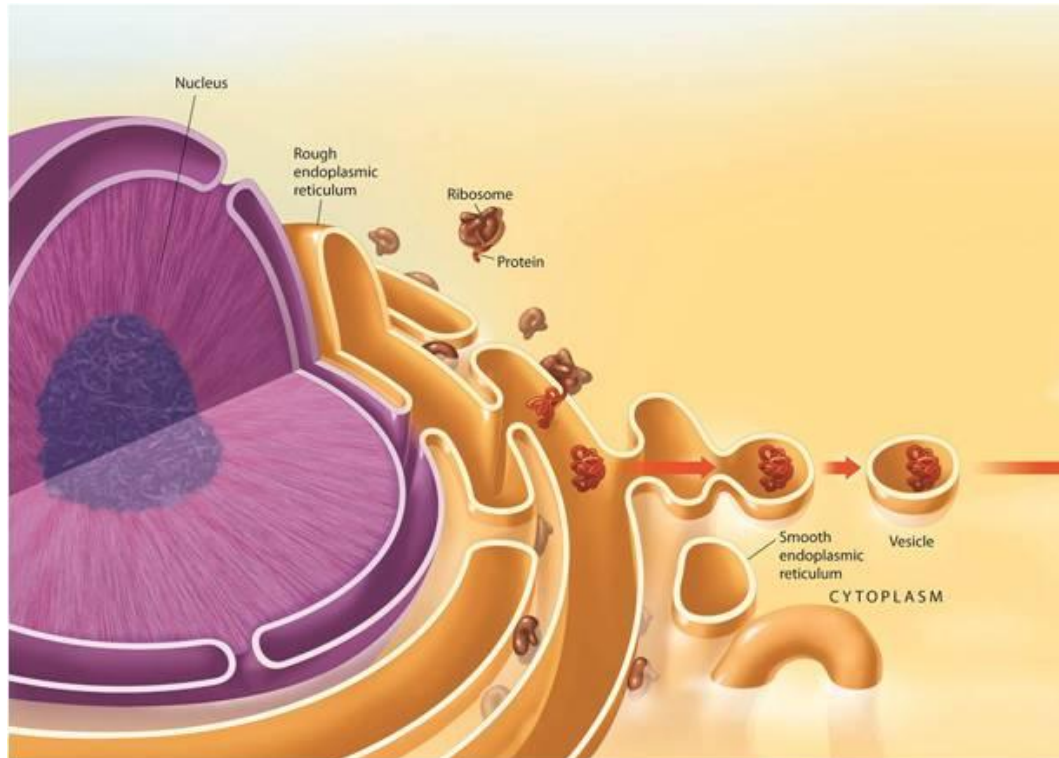
Noctiluca - "Sea Sparkle"

Noctiluca are single celled organisms that live in colonies together. Although they have chloroplasts, they also hunt prey. Some can also show bioluminescence that lights up at night. One variety is responsible for the toxic red tide seen at certain times of the year.



Station 6: Organelles that build proteins

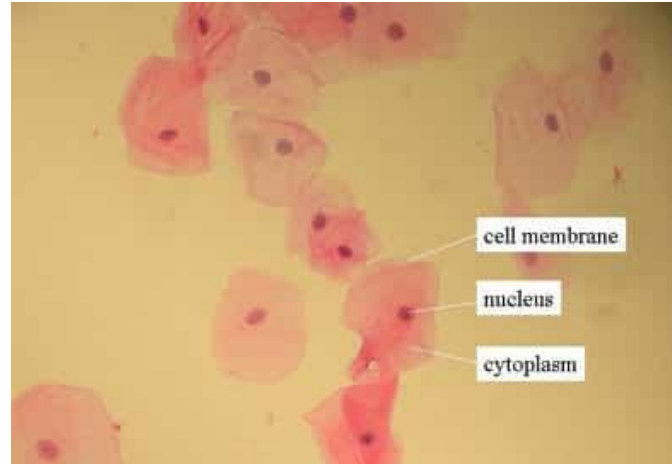
The MAJOR purpose of organelles is to work together to make the proteins that help life function. The steps to making a protein is depicted in the diagram below. Use this diagram to summarize the steps to protein synthesis and to answer the questions that follow.



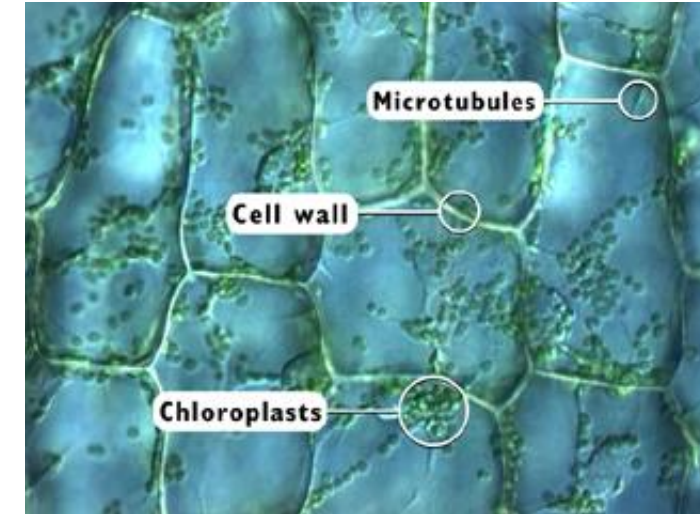
Station 7: Identifying Cells

Cells are diverse in their structure and function. Using the microscopes at this station identify each cell as either prokaryotic or eukaryotic. If the cell is Eukaryotic, also distinguish if it is a plant or animal cell. Explain why.

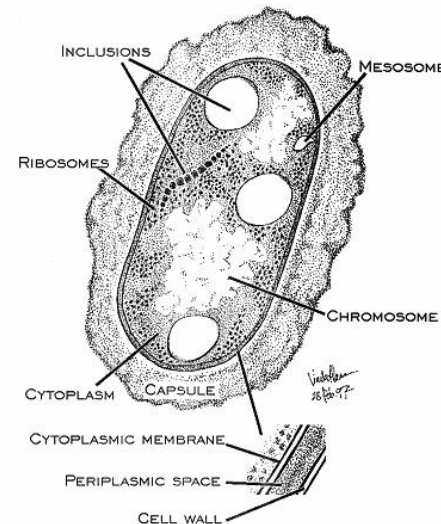
Cell 1



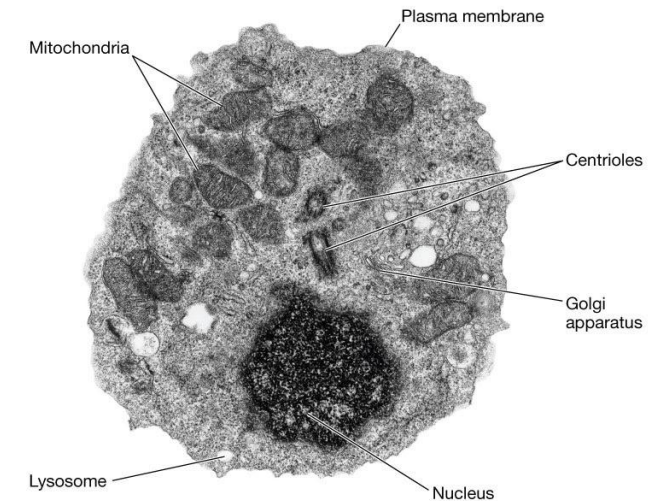
Cell 2



Cell 3



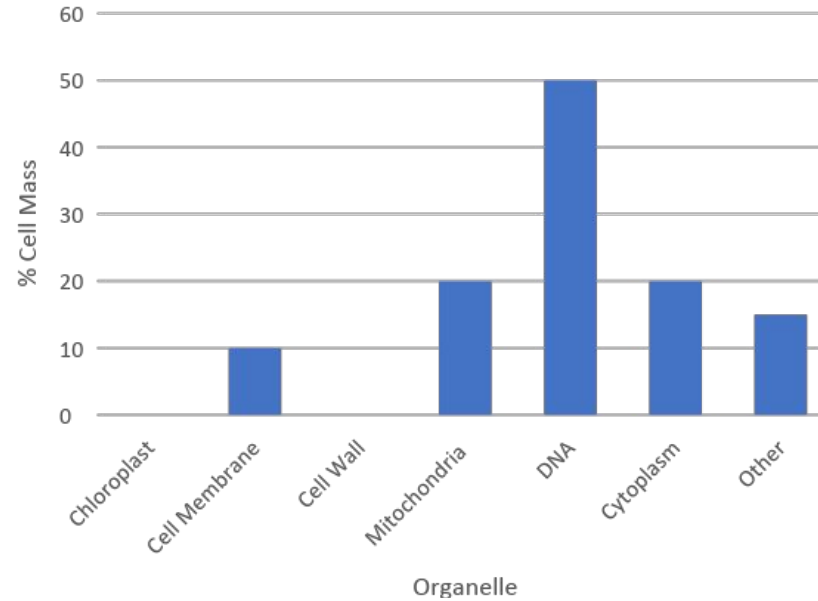
Cell 4



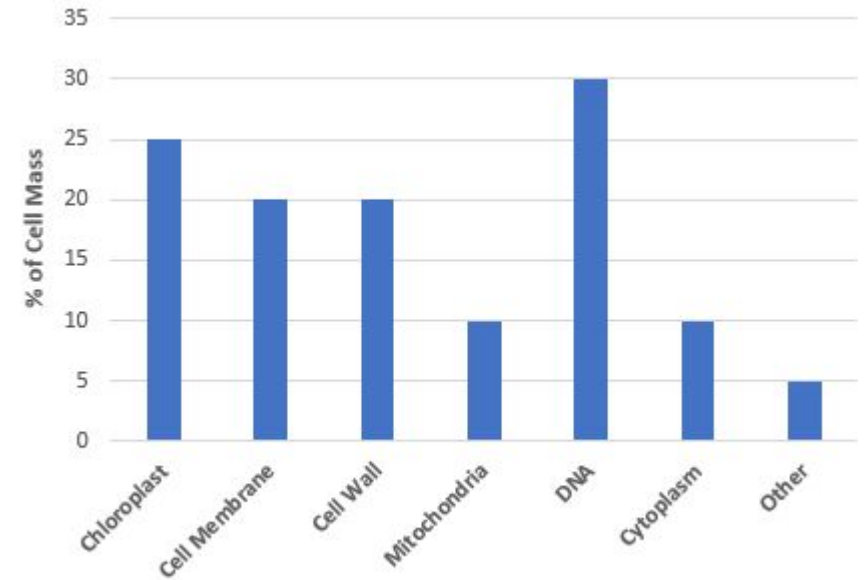
Station 7: Identifying Cells Cont.

Cells are diverse in their structure and function. Based on the graphs shown, identify if the cell is Eukaryotic, also distinguish if it is a plant or animal cell. Explain why.

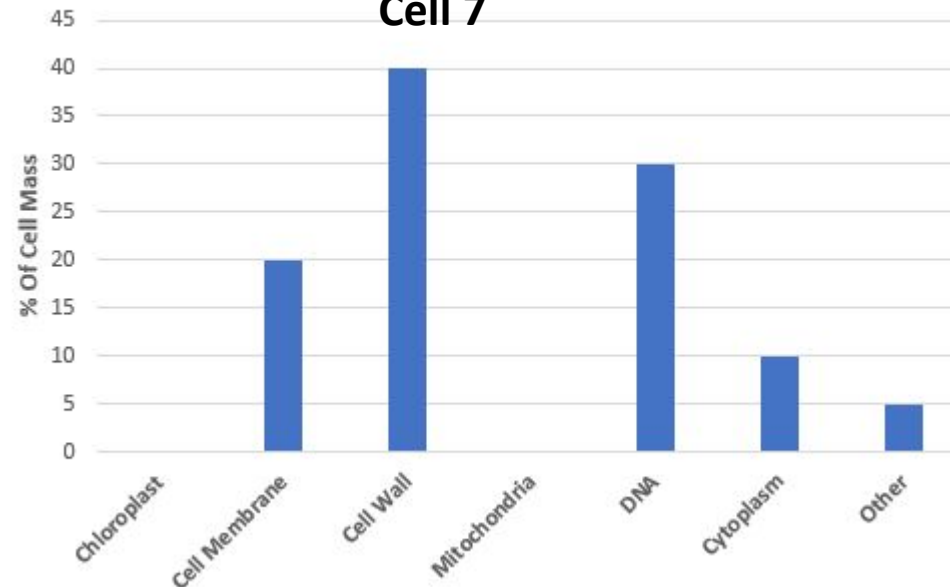
Cell 5

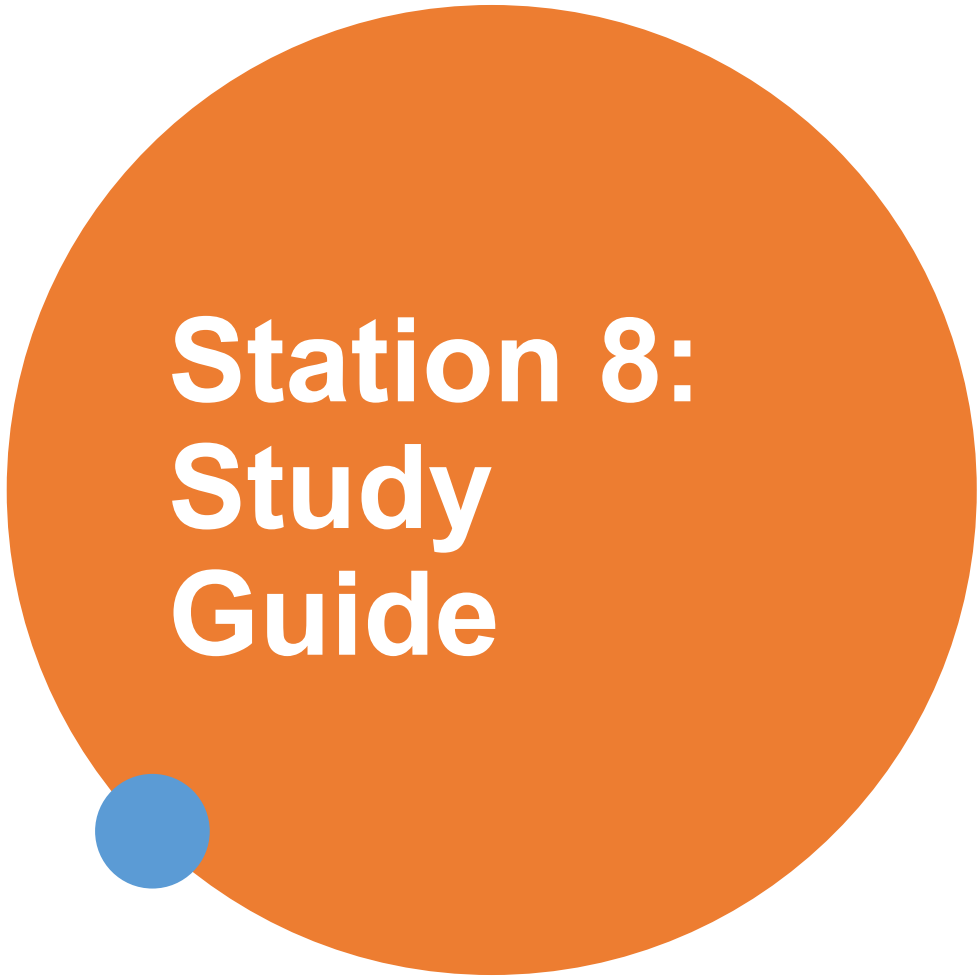


Cell 6



Cell 7





Station 8: Study Guide



Work on your study guide, quiz on vocab, and/or complete any other incomplete packet practice pages with your group to help you prepare for your test!